Estimated Position of the Moving Axes of a Mechanical Arm from an Unknown State

Eduardo M. de C. Caixeta*. Alisson Assis Cardoso.**
Renato França de Almeida***

Federal University of Goiás (UFG), Goiânia, Goiás, Brazil
**(e-mail: eduardocaixeta@discente.ufg.br).*
*** (e-mail: alsnac@ufg.br)*
**** (e-mail: renatofrn@gmail.com)*

Abstract: This work presents an improved methodology for monocular robot pose estimation by enhancing a holistic pose estimation framework through the integration of depth-aware feature extraction and self-supervised learning. The proposed approach incorporates a depth estimation network to extract complementary geometric information from RGB images, generating enriched feature representations through depth maps and heatmaps. These features are concatenated and compressed before being used in the original pose estimation architecture, preserving the network structure while improving representation quality. The training process consists of two stages. First, the model is trained with synthetic data to learn geometric relationships between robot joints, camera parameters, and 3D keypoints. Then, the model is refined using self-supervised learning with real data, guided by kinematic and segmentation consistency constraints instead of ground-truth labels. The self-supervised strategy uses forward kinematics reconstruction and differentiable rendering to maintain geometric consistency between predicted robot pose and image observations. Experimental results show that the proposed method achieves competitive performance in robot pose estimation tasks, slightly outperforming previous methods while maintaining stable training and strong generalization to real-world scenarios. These results indicate that combining synthetic pretraining, depth-enhanced features, and self-supervised adaptation is effective for improving monocular robot pose estimation.

Resumo: O presente estudo propõe uma metodologia avançada voltada para a estimativa de pose de robôs monoculares. O método aprimora uma estrutura holística já existente por meio da integração de aprendizado autossupervisionado e extração de características conscientes de profundidade. Para isso, uma rede de estimativa de profundidade é incorporada ao sistema com o intuito de obter informações geométricas complementares das imagens RGB, gerando representações ricas por meio de mapas de calor e de profundidade. Essas características são compactadas e concatenadas, o que melhora a qualidade dos dados sem alterar a arquitetura original da rede. O treinamento é estruturado em duas etapas principais. Na primeira fase, o modelo utiliza dados sintéticos para compreender as relações geométricas entre os parâmetros da câmera, as articulações do robô e os pontos-chave 3D. Na segunda fase, ocorre o refinamento com dados reais por meio de aprendizado autossupervisionado. Esse processo utiliza a renderização diferenciável e a reconstrução por cinemática direta para manter a consistência entre as observações da imagem e a pose prevista, dispensando o uso de rótulos de ground-truth. Os resultados experimentais revelam que a abordagem atinge um desempenho competitivo, superando métodos anteriores. Além disso, destaca-se a forte capacidade de generalização para cenários reais e a estabilidade durante o treinamento. Tais resultados evidenciam que a união entre o pré-treinamento sintético, a adaptação autossupervisionada e as características aprimoradas por profundidade é uma estratégia eficaz para a estimativa de pose de robôs.

Keywords: Robot Pose Estimation; Monocular Depth Estimation; Computer Vision; Deep Learning; Convolutional Neural Networks; Keypoint Detection Depth-Aware Feature Fusion.

1. INTRODUCTION

In recent years, Artificial Intelligence (AI) has become an essential tool in daily activities across society and industry (Pratti, 2025). Among its research areas, computer vision has stood out as a key driver of technological advances, particularly in healthcare, automation, and robotics (Barbosa, 2023).

In robotics, several recent studies focus on estimating robotic manipulator pose using RGB images combined with joint state information. Methods such as CtRNet-X and RoboPose have demonstrated high accuracy in pose estimation tasks (Labbé, 2021; Lu, 2024). However, the need for prior knowledge of joint angles limits these approaches, reducing their applicability in scenarios where such data is unavailable or unreliable (Lu, 2023). As a result, there is an increasing

demand for models capable of accurately estimating pose using only visual information (Ban, 2024).

Within this line of research, RoboPose (Labbé, 2021) formulates pose estimation as an iterative render-and-compare process, while CtRNet (Lu, 2023) introduces a self-supervised sim-to-real strategy based on foreground segmentation and differentiable rendering. More recently, CtRNet-X (Lu, 2024) extended this framework to partially visible manipulators in real-world conditions, and the Holistic Robot Pose Estimation (HRPE) framework (Ban, 2024) achieved state-of-the-art results on the DREAM benchmark (Lee, 2020) by jointly estimating the camera-to-robot rotation, the translation and the joint states directly from a single RGB image. Despite these advances, such models still rely on large amounts of synthetic training data, which increases computational cost and training time.

In this context, this research investigates the use of Convolutional Neural Networks to estimate the pose of a 7-DOF articulated robotic manipulator (Panda model). Based on the estimated pose, the proposed method also performs robot segmentation in the input image using forward kinematics and camera intrinsic parameters.

The main contributions of this work are: (i) the integration of depth-aware features, extracted by the Depth Anything V2 network, into the HRPE framework without modifying its original architecture; (ii) a two-stage training strategy that combines synthetic pretraining with self-supervised refinement on real data, guided by kinematic and segmentation consistency constraints; and (iii) the demonstration that the synthetic training set can be reduced from more than 26,000 to only 8,000 samples (a reduction of approximately 70%) while slightly outperforming the baseline in accuracy. The remainder of this paper is organized as follows: Section 2 presents the methodology, Section 3 describes the training procedure, Section 4 discusses the results, and Section 5 concludes the work.

2. METHODOLOGY

2.1 Intrinsic and Extrinsic Matrices

Among the various techniques and models employed in computer vision, the projection of 3D points onto image coordinates plays a central role in keypoint estimation using a single RGB image as input.

Within the mathematical models available in the literature, the pinhole camera model stands out due to its simplicity and effectiveness. In this model, a point in the world coordinate system.

$$W = (X, Y, Z)^T \tag{1}$$

is mapped onto the image plane as

$$M = \left(\frac{fX}{Z}, \frac{fY}{Z}, Z\right)^T \tag{2}$$

according to the following projection relationship:

$$M = \mathbf{K}\,[\mathbf{R} \mid \mathbf{t}]\,W \tag{3}$$

where W is the 3D point in homogeneous world coordinates, M is its projection onto the image plane, and R and t are the rotation and translation relating the world frame to the camera frame. The matrix **K** represents the intrinsic camera matrix, which encapsulates the internal parameters of the camera, such as focal lengths and the coordinates of the principal point:

$$\mathbf{K} = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix} \tag{4}$$

In this matrix:

- f_x: denotes the focal length along the x-axis;
- f_y: denotes the focal length along the y-axis;
- c_x: represents the x-coordinate of the principal point;
- c_y: represents the y-coordinate of the principal point.

The matrices **R** and **t** compose the extrinsic parameters, which define the rotation and translation of the camera with respect to the world coordinate system, respectively (Waleed, 2024). To ensure a geometrically consistent projection, the rotation matrix satisfies $\mathbf{R} \in SO(3)$;

2.2 Depth Anything

Obtaining depth information directly from RGB-D cameras is not always viable due to hardware limitations or high equipment costs, which makes depth data unavailable in many real-world scenarios. To address this issue, monocular depth estimation methods have been proposed, with Depth Anything V2.0 standing out as a state-of-the-art approach for estimating depth from single RGB images.

Depth Anything V2.0 is trained on more than 60 million images using a Teacher–Student learning framework. The model uses DINOv2 as a feature extractor, whose outputs are processed by the DPTHead module to refine features and generate the final depth map through convolution and interpolation operations (Yang, 2024).

In this framework, a teacher model based on DINOv2-Giant is first trained using labeled synthetic data. This model then generates pseudo-labels for real-world images, which are used to train multiple student models based on DINOv2-Small, Base, and Large. This strategy enables knowledge transfer from synthetic data to real-world scenarios, improving depth estimation performance (Yang, 2024).

2.3 HRPE Model

Currently, the Holistic Robot Pose Estimation (HRPE) model (Ban, 2024) has the best metrics in the literature when we talk about the Deep Robot-to-Camera Extrinsics for Articulated Manipulators - DREAM dataset (Lee, 2020), whose main objective is to provide a robust and diverse dataset for estimating the pose of a robotic arm from RGB images.

Originally, the model consists of 4 distinct neural networks: DepthNet, JointNet, RotationNet, and KeypointNet, as shown in Figure 1.

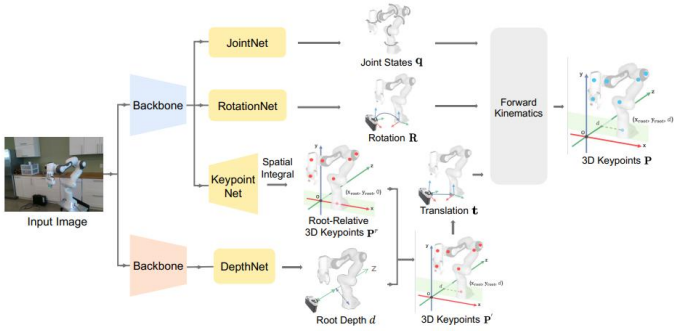


Fig. 1. General structure of the Holistic Robot Pose Estimation (HRPE) model. Adapted from (Ban, 2024).

Starting from an RGB image I, we use each module of this network set to perform specific tasks, which are:

DepthNet: Estimate the absolute depth d of the root;

JointNet: Retrieve joint status parameters **q**;

RotationNet: Predicts the camera rotation matrix **R** relative to the robot;

KeypointNet: Calculate 3D positions relative to the root of the key points $\mathbf{P_r} \in R^{Nx3}$, where *N* corresponds to the total number of joints present in the robotic arm.

Subsequently, the outputs from DepthNet and KeypointNet are combined to determine the absolute keypoint positions $\mathbf{P}' \in R^{Nx3}$ which are used to compute the translation vector t. Finally, the joint parameters q, along with R and t, are applied through a forward kinematics process to obtain the final 3D keypoint estimates $\mathbf{P} \in R^{Nx3}$, (Ban, 2024). In this forward kinematics step, the kinematic chain described in the robot URDF model is traversed: each joint angle in q defines a homogeneous transformation between consecutive links, and the successive composition of these transformations from the base frame yields the 3D position of each joint, which is finally expressed in the camera frame through R and t.

The JointNet, RotationNet, and KeypointNet networks share the same ResNet-50 feature extractor, while the DepthNet network uses the more robust HRNet-32 extractor.

2.3.1 DepthNet

The DepthNet network is a single-layer convolutional network that seeks to estimate the distance between the base of the arm and the camera. However, the estimation of the depth of monocular objects suffers from significant ambiguities, since the absolute depth value and the camera's intrinsic values are deeply intertwined. Therefore, the use of DepthNet is divided into two parts. Initially, using the camera's intrinsic data, we calculate a normalized depth value (Ban, 2024).

$$\mathbf{d} = \sqrt{\frac{fx \cdot fy \cdot A_{real}}{A_{bbox}}} \tag{5}$$

In this equation, A_{real} represents the robot area projected onto the camera x–y plane in mm², while A_{bbox} corresponds to the area of the robot bounding box in pixels².

Thus, we use the convolutional network only to estimate a correction factor λ from the features f_1, which, once multiplied by d, can provide the robot's absolute root depth. In this formulation, λ is a dimensionless scalar learned directly from the image features, whose role is to compensate for the scale ambiguity inherent to monocular depth estimation, as expressed in (6).

$$pred_{depth} = \lambda \cdot \mathbf{d} \tag{6}$$

2.3.2 JointNet

Through several MLP blocks, we use the extracted features $\mathbf{f_2}$ to estimate the angle in radians of each joint of the arm (Ban, 2024).

2.3.3 RotationNet

Similarly, in RotationNet we aim to estimate the camera-to-robot rotation matrix through MLP blocks using the same features f_2 (Ban, 2024).

2.3.4 KeypointNet

The objective of this convolutional network is to estimate the location of the keypoints relative to the root point in an R^{Nx3} space, where N corresponds to the number of robot joints, using transposed convolutional networks.

2.4 New Model

The methodology adopted to improve the HRPE model (Ban, 2024) consists of incorporating the Depth Anything network (Yang, 2024) in order to explore depth information and generate heatmaps from the images, enriching the feature representation.

Thus, we concatenate and compress the new features and use them to train the model while maintaining the network architecture, as shown in Fig. 2. The left side of Fig. 2 depicts the baseline HRPE pipeline, in which the HRNet-32 and ResNet-50 backbones process only the RGB image, whereas the right side details the proposed configuration, in which the Depth Anything module operates in parallel and its depth features are fused into the HRNet branch through an explicit depth integration module before the pose regression heads.

In addition, we seek to optimize the model's performance by modifying its hyperparameters, training approach, and reducing the amount of data during training with synthetic values.

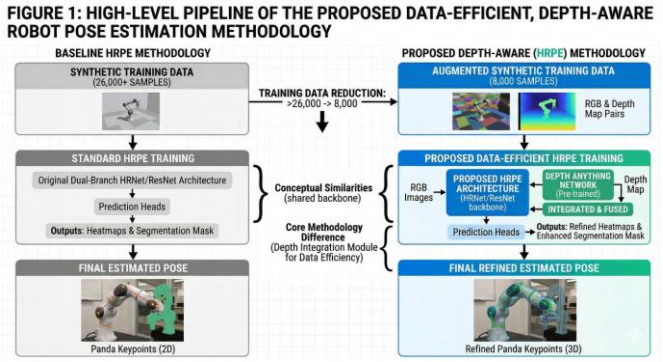


Fig. 2. Detailed Architectural Comparison Showing Explicit Depth Integration. On the left side, the diagram illustrates the baseline Holistic Robot Pose Estimation (HRPE) architecture, which utilizes a shared dual-branch backbone (HRNet-W32 and ResNet-50) to process a monocular RGB image, leading to joint heatmaps, masks, offsets, and the final robot pose. On the right side, the diagram details the proposed depth-enhanced architecture. This configuration introduces a parallel Depth Anything Network module (highlighted in green) and an explicit Depth Integration Module (highlighted by the dashed blue box), where depth features are directly injected via fusion into the HRNet branch to refine spatial understanding.

2.5 Training Loss Functions for Synthetic and Self-Supervised Learning

Because each network is responsible for estimating different variables, the adopted loss functions must account for these differences. Thus, during synthetic training, the loss functions are defined as:

$$L_{depth} = ||d - \hat{d}||_1 \quad (7)$$

$$L_{joint} = ||q - \hat{q}||_2 \quad (8)$$

$$L_{rot} = ||R - \hat{R}||_2 \quad (9)$$

$$L_{trans} = ||t - \hat{t}||_2 \quad (10)$$

$$L_{tkps} = ||P - \hat{P}||_2 + ||p - \hat{p}||_2 \quad (11)$$

$$L'_{kps} = ||P' - \hat{P}'||_2 + ||p' - \hat{p}'||_2 \quad (12)$$

Where $\mathbf{d}$, $\mathbf{q}$, $\mathbf{R}$, $\mathbf{t}$, $\mathbf{P}$ and $\mathbf{P}'$ represent, respectively, the reference keypoint depth, joint position values, camera rotation matrix, camera translation, keypoint coordinates based on the robot forward kinematics, and their regressed locations. Additionally, $\mathbf{p}$ and $\mathbf{p}'$ represent the 2D projections of points $\mathbf{P}$ and $\mathbf{P}'$ onto the image plane through the camera intrinsic matrix (Ban, 2024).

From these partial loss functions, the total synthetic training loss is defined as:

$$L = L_{depth} + L_{joint} + L_{rot} + L_{trans} + L_{tkps} + L'_{kps} \quad (13)$$

For real data, training is performed using a self-supervised approach, meaning that no ground-truth labels are used. Instead, learning is driven by consistency constraints imposed by the robot's kinematic structure and the visual information present in the image, (Ban, 2024). Let $\hat{\mathbf{P}}$ be the keypoint locations obtained from forward kinematics while respecting the robot structure, and $\hat{\mathbf{P}}'$ be the values predicted by the model. The loss function for these variables is defined as:

$$L_{3D} = ||P' - \hat{P}'||_2 \quad (14)$$

Using the values predicted by the model, the robot geometric description available in the Unified Robot Description Format (URDF) file is used to reconstruct its configuration through forward kinematics. Then, differential rendering is applied to obtain a binary segmentation mask M_{render}. Additionally, the CtRNet segmentation model (Lu, 2023) is used to obtain another segmentation mask M_{seg}. The mask loss function is computed using the Intersection over Union (IoU) dissimilarity metric (Ban, 2024):

$$L_{mask} = 1 - \frac{S_{intersection}}{S_{union}} \quad (15)$$

Therefore, the total loss for this training stage is defined as:

$$L = L_{3D} + L_{mask} \quad (16)$$

3. TRAINING

In all experiments, the model was trained on the DREAM dataset (RealSense split) using a Google Colab environment equipped with an NVIDIA L4 GPU. Initially, DepthNet was trained for 500 epochs (learning rate of 1×10^{-4}) to initialize the full architecture, which was then trained on synthetic data. A key distinction of our approach is the drastic reduction of this synthetic dataset to only 8,000 images, significantly decreasing computational costs while maintaining competitive performance.

The best-performing model subsequently underwent self-supervised fine-tuning on real-world data for 400 epochs (learning rate of 1×10^{-5}).

This stage relied on robot kinematic structure and visual consistency constraints rather than ground-truth labels. Finally, training diagnostics (Figure 3) demonstrate stable convergence and predictive stability without severe overfitting, confirming the robustness of the adopted strategy.

Table 1. Training Configuration and Hyperparameters I.

Training Step	Model	Data Type	Learning Rate
Step 1	DepthNet	Synthetic	1×10^{-4}
Step 2	Full Model	Synthetic	1×10^{-5}
Step 3	Full Model	Real	1×10^{-5}

Table 2. Training Configuration and Hyperparameters II.

Training Step	Epochs	Batch	Optimizer
Step 1	500	32	Adam
Step 2	400	16	Adam
Step 3	400	16	Adam

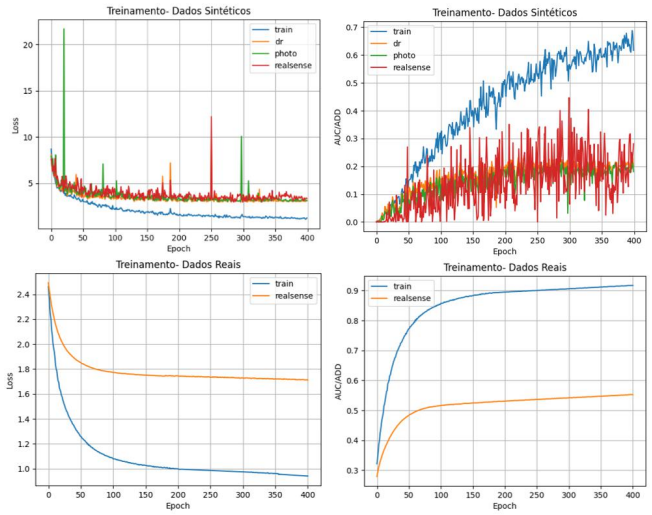


Fig. 3. The figure presents a comprehensive view of the learning dynamics across different data domains. The top row illustrates the metrics during the synthetic data training phase, whereas the bottom row displays the corresponding metrics for the real data phase. For both rows, the left column exhibits the loss curves over the epochs, demonstrating the model's convergence behavior. The right column presents the progression of the Area Under the Curve for the Average Distance of Distinguishable points (AUC/ADD), characterizing the predictive stability and accuracy improvements throughout the process.

4. RESULTS

As illustrated in Figure 4, the proposed method achieves reliable monocular robot pose estimation, yielding a 75.4% keypoint prediction accuracy at the standard 60 mm threshold (Ban, 2024). This slightly outperforms existing methods such as RoboPose (Labbé, 2021), with 74.3%, and the baseline HRPE (Ban, 2024), with 75.2%, as summarized in Table 3. Crucially, this competitive performance is achieved while drastically reducing data dependency — using only 8,000 synthetic training samples compared to the standard >26,000. These results confirm that integrating depth-aware features effectively maintains real-world robustness while significantly enhancing data efficiency.

The evaluation relies on the Area Under the Curve of the Average Distance (AUC/ADD) metric, which is the standard protocol of the DREAM benchmark and therefore allows direct comparison with previously published results (Lee, 2020; Ban, 2024). Since ADD measures the average Euclidean distance between estimated and ground-truth 3D keypoints, it jointly captures rotation, translation and joint-state errors in a single value; nevertheless, the use of complementary metrics, such as the 2D percentage of correct keypoints, is planned as future work to provide a more granular error analysis.

Regarding computational cost, reducing the synthetic training set from more than 26,000 to 8,000 samples decreases the number of training iterations per epoch by approximately 70%, proportionally reducing GPU usage during the synthetic pretraining stage. The additional overhead introduced by the Depth Anything V2 module is restricted to feature extraction, since its weights remain frozen during training.

Table 3. Comparison with state-of-the-art methods on the DREAM RealSense dataset (accuracy at the 60 mm ADD threshold).

Method	Synthetic Samples	Accuracy (%)
RoboPose (Labbé, 2021)	–	74.3
HRPE (Ban, 2024)	>26,000	75.2
Proposed method	8,000	75.4

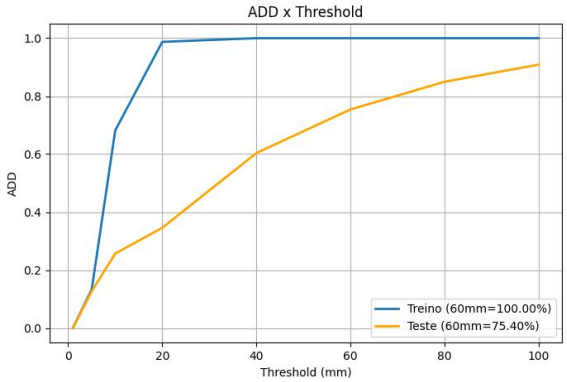


Fig. 4. The blue curve represents the model's performance on the synthetic training dataset, while the orange curve demonstrates the zero-shot inference performance on the real-world test dataset. At the standard 60 mm threshold, the proposed method successfully achieves a 75.40% accuracy on real data, demonstrating robust sim-to-real transferability despite the significantly reduced synthetic training set.

5. CONCLUSION

The results demonstrate that the proposed method achieves competitive performance in monocular robot pose estimation, providing reliable spatial localization of robot joints and slightly outperforming existing approaches. The integration of depth-aware features extracted from monocular RGB images improves the model's ability to learn geometric relationships between visual data and robot pose parameters, contributing to robust performance in real-world scenarios.

Linking the objectives of this study to the results obtained: (i) the integration of depth-aware features from the Depth Anything network proved effective, since the enriched representations preserved accuracy on real data; (ii) the two-stage training strategy provided stable convergence without severe overfitting, as shown by the training diagnostics; and (iii) the reduction of the synthetic dataset from more than 26,000 to 8,000 samples was achieved with a slight improvement in accuracy (75.4% against 75.2% of the baseline), lowering computational cost and training time.

As future work, we intend to validate the proposed approach on additional datasets and robotic platforms beyond the DREAM RealSense split, to evaluate complementary metrics such as the 2D percentage of correct keypoints and the per-joint angular error, and to investigate the deployment of the model in real-time applications. The main limitations of the current self-supervised stage, namely its dependence on reliable segmentation masks and on an accurate URDF description of the robot, will also be addressed.

ACKNOWLEDGMENTS

The authors thank FUNAPE – Research Support Foundation (CNPJ No. 00.799.205/0001-89) for financial support through the scholarship granted for the project "Implementation of CERISE – Center of Excellence in Wireless Intelligent Networks and Advanced Services" (cost center 08.193), carried out at the Federal University of Goiás – UFG and coordinated by Flavio Henrique Teles Vieira.

REFERENCES

Ban, S., Fan, J., Ma, X., Zhu, W., Qiao, Y., and Wang, Y. (2024). Real-time holistic robot pose estimation with unknown states. *arXiv preprint arXiv:2402.05655*. https://arxiv.org/abs/2402.05655. Last accessed 2025/09/20

Barbosa, D.M., Amorim, T.A., and Mascarenhas, A.P.D.M. (2023). Aprendizado de máquina no campo da visão computacional para uso na robótica inteligente. *CINTERGEO*, UNEB.

Labbé, Y., Carpentier, J., Aubry, M., and Sivic, J. (2021). Single-view robot pose and joint angle estimation via render & compare. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, Nashville.

Lee, T.E., Tremblay, J., To, T., Cheng, J., Mosier, T., Kroemer, O., Fox, D., and Birchfield, S. (2020). Camera-to-robot pose estimation from a single image. *arXiv preprint arXiv:1911.09231*. https://arxiv.org/abs/1911.09231. Last accessed 2025/11/12

Lu, J., Richter, F., and Yip, M.C. (2023). Markerless camera-to-robot pose estimation via self-supervised sim-to-real transfer. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). IEEE, Vancouver.

Lu, J., Zekai, L., Xie, T., et al. (2024). CtRNet-X: Camera-to-robot pose estimation in real-world conditions using a single camera. *arXiv preprint arXiv:2409.10441*. https://arxiv.org/abs/2409.10441. Last accessed 2025/10/18

Pratti, W.L. (2025). Inteligência artificial: definições desta e seu uso na educação à distância. *Revista Educação Contemporânea* 2(3), 1–15.

Waleed, M., Rauf, A., and Taj, M. (2024). Camera calibration through geometric constraints from rotation and projection matrices. arXiv preprint arXiv:2402.08437. https://arxiv.org/abs/2402.08437. Last accessed 2026/08/01.

Yang, L., Kang, B., Huang, Z., Zhao, Z., Xu, X., Feng, J., and Zhao, H. (2024). Depth Anything V2. *arXiv preprint arXiv:2406.09414*. https://arxiv.org/abs/2406.09414. Last accessed 2026/01/16.